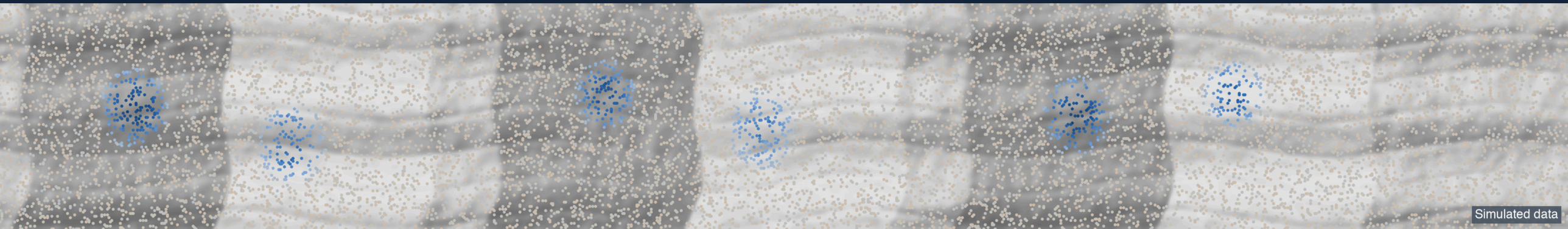
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Landslide Monitoring with InSAR

Measuring slow slope movements from space

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use satellite radar to find slopes that are moving, measure how fast, and judge when that matters.

Which landslides InSAR can see, how slope direction changes what it measures, and how engineers use the results.

THIS LECTURE

Outline

01 **Landslides and why we monitor them**
Types, speeds, and what InSAR adds

02 **Seeing slopes from space**
Line of sight, slope direction, layover, shadow and the speed limit

03 **Techniques and processing**
Point targets, small baselines, offset tracking and the open tools

04 **Reading the results**
Velocity maps, two passes, time series and forecasting failure

05 **Case studies**
Mud Creek, Xinmo and Joshimath, and national ground-motion services

06 **Engineering practice**
Uses, field instruments, limits and exercises

01

PART 1

Landslides and why we monitor them

Types, speeds, and what InSAR adds

LANDSLIDES

What moves, and how

Falls

Rock or debris drops from a cliff in seconds. InSAR can only show where blocks are loosening.

InSAR: source areas only

Topples

Rock columns rotate forward out of a slope. Slow tilting before failure is sometimes visible.

InSAR: sometimes

Slides

A mass moves on a slip surface, rotational or translational. Often slow for years.

InSAR: main target

Spreads

Lateral extension of a slope on a weak layer below. Usually slow.

InSAR: yes

Flows

Debris flows are rapid and cannot be measured; earthflows creep slowly and can.

InSAR: earthflows only

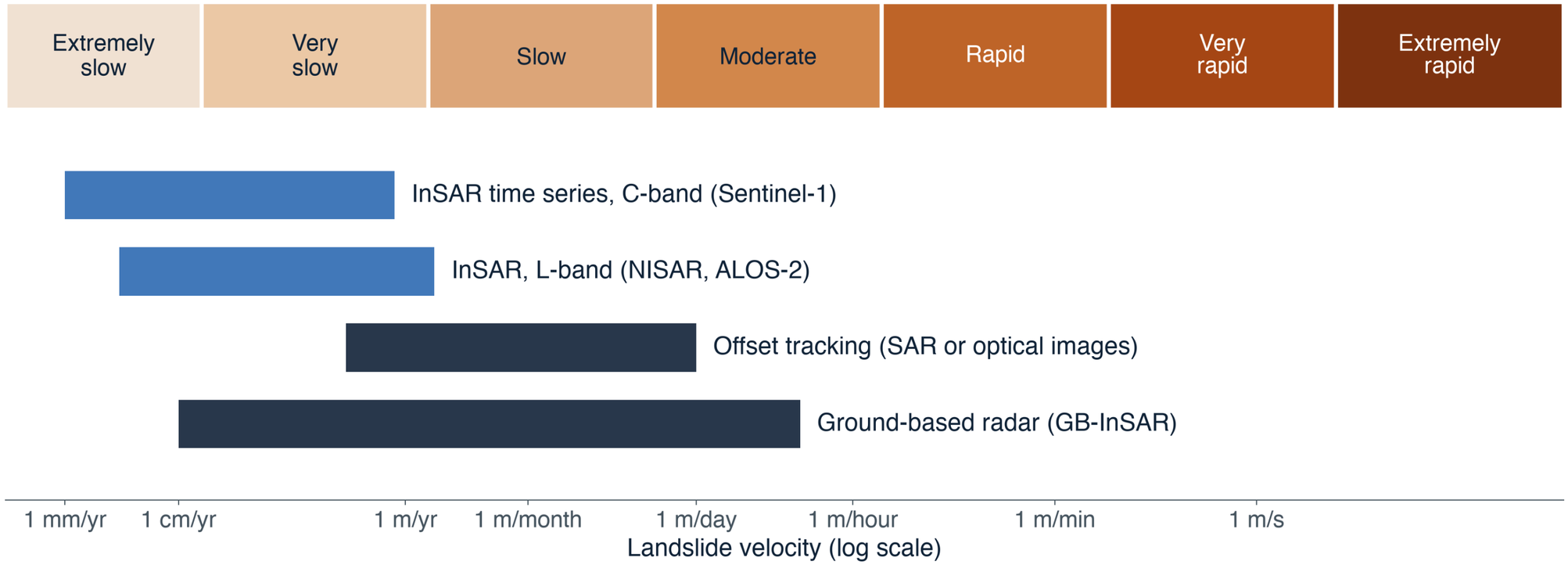
Slope creep

Whole hillsides deforming by millimetres to centimetres a year, often deep-seated.

InSAR: ideal

LANDSLIDES

Landslide speeds, and where InSAR fits



Velocity classes after Cruden and Varnes (1996). Technique ranges are indicative: they depend on the sensor, the revisit time and the site.

LANDSLIDES

Why use InSAR for landslides?

~1 mm/yr
Velocity precision of a good time series

250 km
Width of one Sentinel-1 image

6–12 days
Time between Sentinel-1 images

Since 2014
Free archive to look back in time

Finds landslides nobody knew about

Whole valleys and road corridors are mapped at once, so slow, hidden movement shows up before damage does.

Tells you where to put instruments

Field sensors are expensive. InSAR shows which slopes are moving, so boreholes and GNSS go where they matter.

But only part of the story

Slow movement only, one direction only, and mountains are hard for a side-looking radar. The next part explains why.

02

PART 2

Seeing slopes from space

Line of sight, slope direction, layover, shadow and the speed limit

GEOMETRY

InSAR measures only one direction

$$v(LOS) = C \times v(slope)$$

The radar measures only the part of the motion along its line of sight (LOS). C is the cosine of the angle between the line of sight and the downslope direction.

C near ±1

Motion is along the line of sight: almost all of it is measured.

C near 0

Motion is across the line of sight: almost nothing is measured.

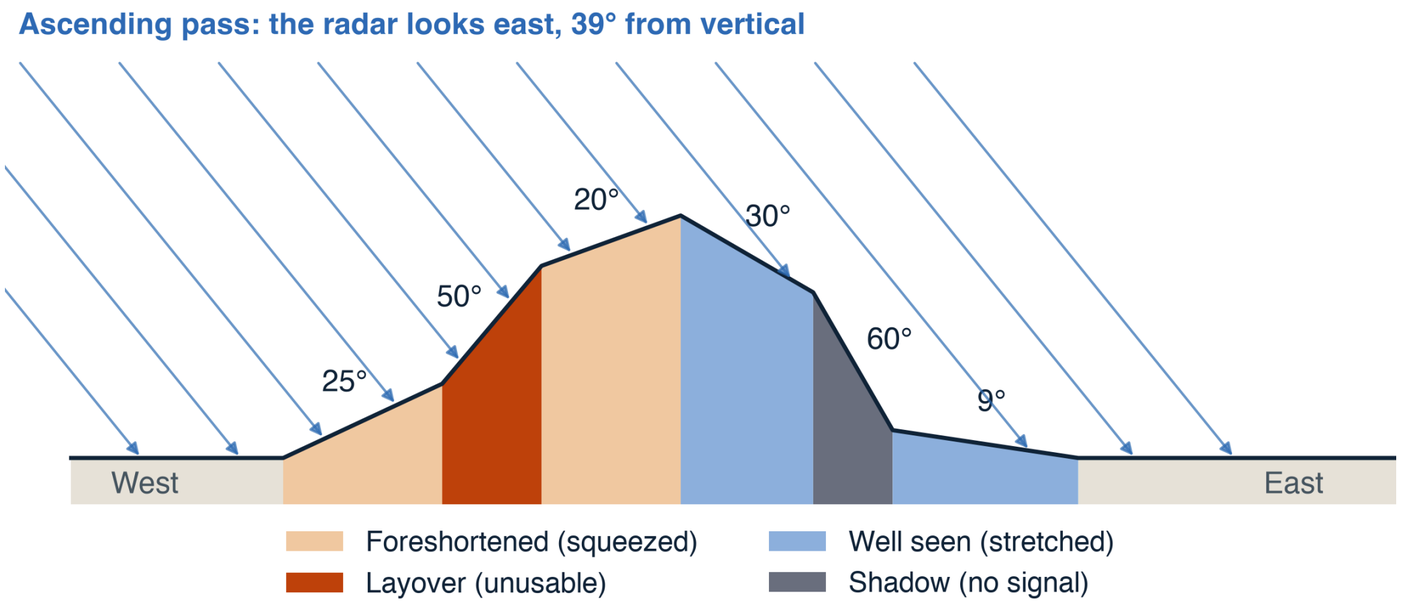
C depends on

- The direction the slope faces (aspect)
- How steep the slope is
- The pass: ascending looks east, descending looks west
- The incidence angle (about 30–46° for Sentinel-1)

Both passes fly close to north–south, so motion towards north or south is hard to see from either.

GEOMETRY

Mountains are hard for a side-looking radar



Ascending pass over a ridge; slope angles are marked.

Layover: a slope facing the radar that is steeper than the incidence angle (about 39°). Unusable.

Shadow: a slope facing away that is steeper than 90° – 39° = 51°. No signal at all.

Foreshortening: slopes facing the radar are squeezed into a few pixels.

Fix: use the other pass, and mask layover and shadow with the elevation model.

GEOMETRY

Which slopes does each pass see?

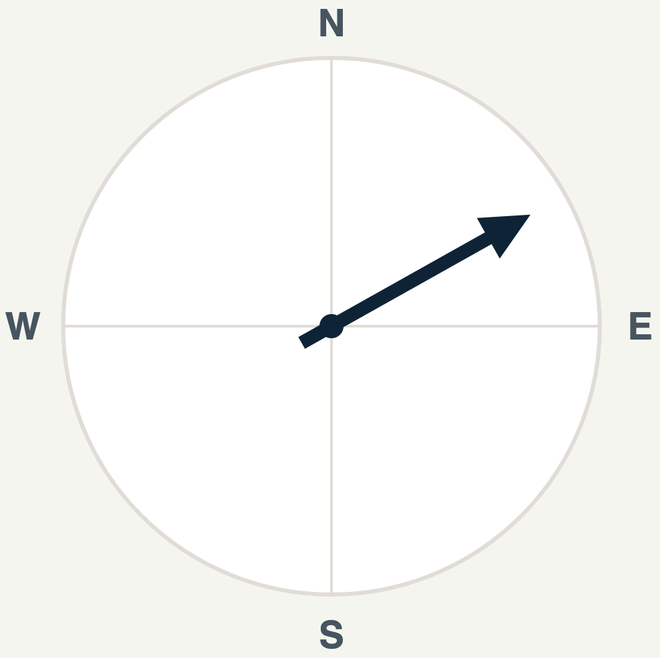


Downslope motion seen: 30° slope, 39° incidence.

- East-facing slope: ascending sees **92%**, descending only **15%**.
- West-facing slope: the reverse, 92% from descending and 15% from ascending.
- North- or south-facing slopes: only 29–48% from either pass, mostly the sinking part of the motion.
- Always process both passes, then use the better one for each slope.

GEOMETRY

Turn the slope: what does each pass see?



Ascending pass looks east

Descending pass looks west

The slope faces

North-east (61°)

Arrow = direction of downslope motion

Ascending pass

90%

Seen well: moving away from the satellite

Descending pass

3%

Poorly seen: use the other pass

Slope 30°, incidence 39°. % = share of the downslope motion along the line of sight.

GEOMETRY

From line of sight to downslope velocity

$$v(\text{slope}) = v(\text{LOS}) \div C$$

01 Landslide A faces east-north-east (67°) with a 35° slope.

02 From the elevation model, ascending C = −0.95.

03 Measured ascending LOS velocity: −35.6 mm/yr (moving away).

04 Downslope velocity = −35.6 ÷ (−0.95) ≈ **38 mm/yr**.

Only where |C| is large

Dividing by a small C multiplies the noise: with C = 0.15, an error of 2 mm/yr becomes 13 mm/yr. A common rule: do not project where |C| is below about 0.3.

It assumes the direction

Projection assumes the landslide moves straight down the steepest slope. Check that with field evidence or with both passes.

GEOMETRY

The speed limit: phase aliasing

Phase repeats every half wavelength of motion. If neighbouring pixels move apart by more than about **λ/4 between two images**, the phase jumps ambiguously and the motion cannot be recovered.

Satellite	Band and wavelength	Revisit	λ/4 per revisit	Largest velocity difference
Sentinel-1 (two satellites)	C-band, 5.5 cm	6 days	1.4 cm	≈ 0.8 m/yr
Sentinel-1 (one satellite)	C-band, 5.5 cm	12 days	1.4 cm	≈ 0.4 m/yr
NISAR	L-band, 24 cm	12 days	6.0 cm	≈ 1.8 m/yr
ALOS-2	L-band, 23 cm	14 days	5.7 cm	≈ 1.5 m/yr
TerraSAR-X	X-band, 3.1 cm	11 days	0.8 cm	≈ 0.3 m/yr

Faster landslides: shorter revisit, longer wavelength (L-band), or offset tracking.

03

PART 3

Techniques and processing

Point targets, small baselines, offset tracking and the open tools

TECHNIQUES

Choosing a technique

Technique	Best for	Main limit
Differential InSAR (one pair)	Sudden movements, fast first map after an event	Atmosphere; only a snapshot
Persistent scatterers (PS-InSAR)	Towns, roads, bare rock: mm/yr time series on stable point targets	Few points on vegetated slopes
Small baseline (SBAS) and distributed scatterers	Bare soil, sparse vegetation, earthflows	Coarser; needs coherence
Offset tracking	Fast landslides: m/yr to m/day	Precision of decimetres
Corner reflectors	Vegetated slopes with no natural targets	Must be installed and maintained
Ground-based radar (GB-InSAR)	Emergencies and early warning, open-pit mines: images every few minutes	One slope at a time

TECHNIQUES

What reflects radar on a hillside?

Strong and stable

Buildings, roads, retaining walls, bare rock outcrops, corner reflectors. Give persistent scatterers with millimetre precision.

Weak but usable

Bare soil, scree, grass and sparse shrubs. Pixels are averaged as distributed scatterers; coherence drops with time.

No useful signal

Dense forest, crops, snow, water and fresh landslide scars. C-band loses coherence in days; L-band lasts much longer.

TECHNIQUES

A landslide InSAR workflow

01
Choose data

02
Process time series

03
Velocity map and hotspots

04
Check geometry

05
Project to the slope

06
Compare with rainfall

07
Check in the field

08
Report and alert

Data: ascending and descending stacks, at least a year (Sentinel-1, NISAR).

Geometry: compute C, layover and shadow from the elevation model.

Field: walk the hotspots: cracks, tilted trees, damaged structures.

TECHNIQUES

Open tools and ready-made products

EGMS

European Ground Motion Service: free Sentinel-1 point velocities and time series for all of Europe, updated every year.

COMET-LiCSAR and LiCSBAS

Ready-made Sentinel-1 interferograms worldwide, and a free Python tool that turns them into time series.

ASF HyP3 and MintPy

Interferograms processed in the cloud on request, then small-baseline time series in MintPy.

SNAP and StaMPS

The SNAP chain from the practical, exported to StaMPS for persistent-scatterer time series.

ISCE

NASA-JPL processing software; ISCE3 processes NISAR data.

QGIS

Free GIS to map velocities with slope, geology, roads and rainfall.

TECHNIQUES

Mountains add atmosphere and seasons

Atmosphere that looks like motion

Water vapour decreases with height, so the radar delay follows topography. On a slope this can mimic or hide a landslide.

Corrections

Weather models (GACOS, ERA5) or a fit of phase against elevation; long time series also average the noise down.

Seasons break coherence

Snow in winter, fast-growing vegetation in the monsoon. Expect gaps in the time series every year.

Wavelength helps

L-band (24 cm) keeps coherence over vegetation much longer than C-band (5.5 cm).

04

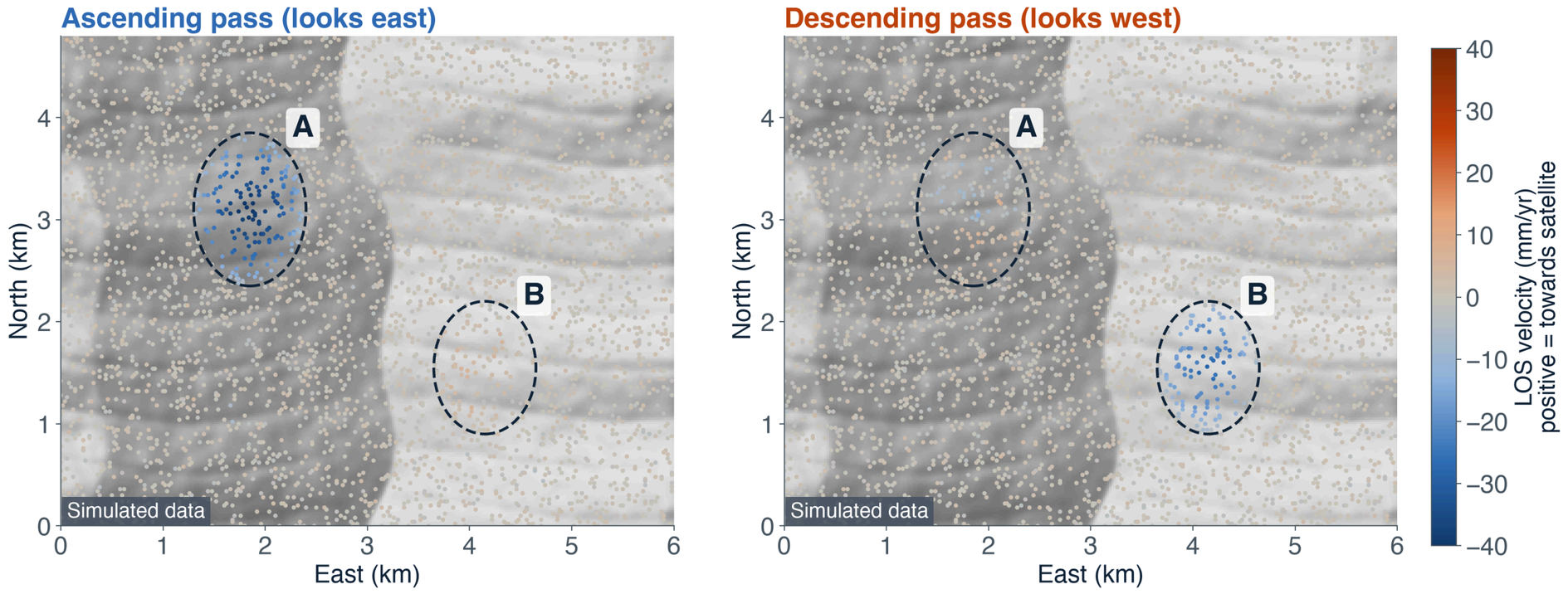
PART 4

Reading the results

Velocity maps, two passes, time series and forecasting failure

RESULTS

One valley, two passes



Simulated example data. Landslide A (east-facing): ascending −36 mm/yr, descending about 0. Landslide B (west-facing): ascending +2, descending −23 mm/yr.

RESULTS

Two passes, two components

Ascending and descending see the same ground from opposite sides. Two measurements let you solve for two components of motion:

Up–down

Vertical motion: sinking at the head, bulging at the toe

East–west

Horizontal motion in the direction of the radar look

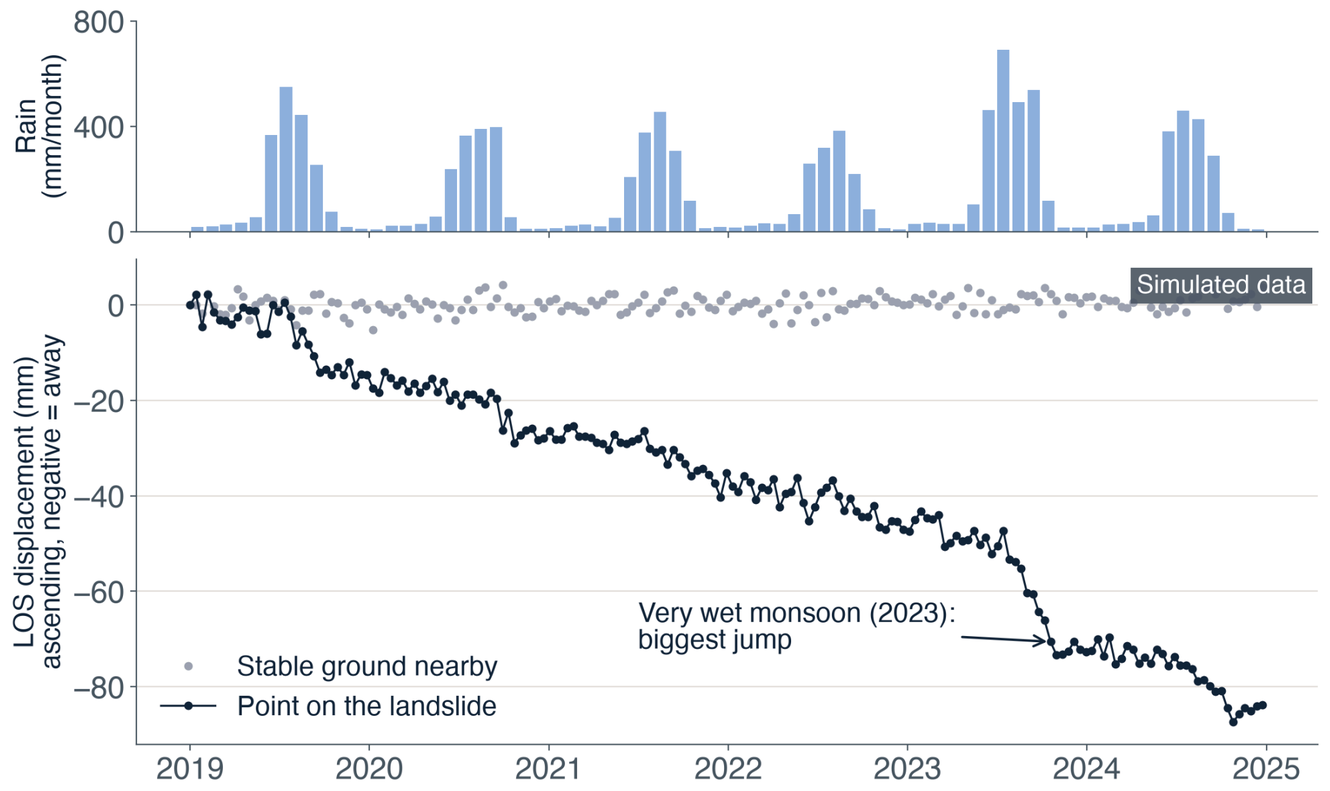
North–south motion stays almost invisible: it is the blind direction of near-polar orbits.

In practice

The points of the two passes are resampled to a common grid first. EGMS provides this as its ‘ortho’ product. For north- or south-facing slopes, assume the motion is downslope instead.

RESULTS

Time series: the landslide breathes with the monsoon



Simulated example data: Sentinel-1 every 12 days, ascending pass.

The landslide moves in steps: fast in the monsoon, almost still in the dry season.

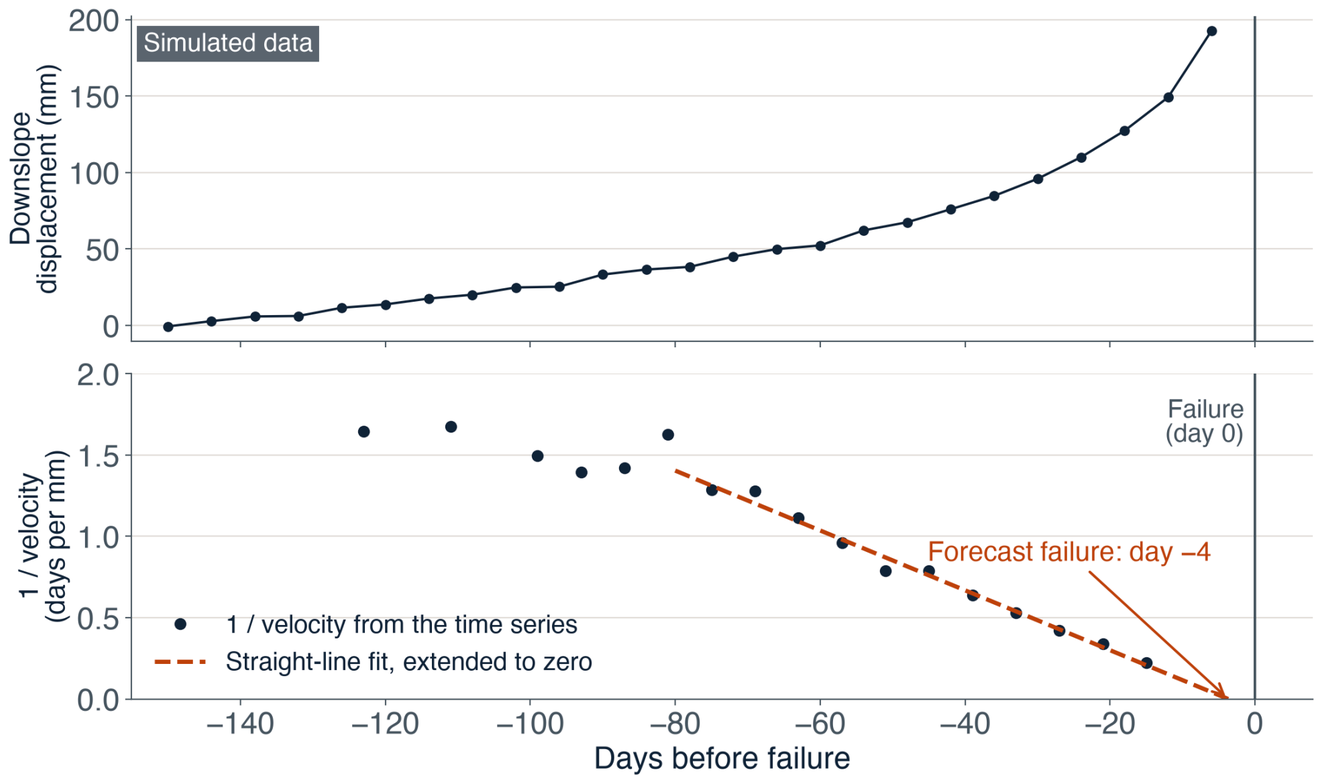
A wetter monsoon (2023 here) gives a bigger step: rainfall is the trigger.

Stable ground nearby stays at zero: it shows the noise level, about ±2 mm.

Watch for a change in the pattern: faster every year, or no slowing in the dry season.

RESULTS

Before failure: the inverse-velocity method



Simulated example data: one image every 6 days, the last one 6 days before failure.

Before many failures the slope accelerates, and **1 / velocity** falls along a straight line (Fukuzono, 1985).

Extend the line to zero: here it forecasts failure on day −4, close to the real day 0.

Needs frequent data: with a 6- to 12-day revisit, the last image can come too late.

RESULTS

Can satellite InSAR give early warning?

Good at

- Finding which slopes are moving, region-wide
- Spotting a slope that is speeding up over months
- Looking back before an event, in the archive
- Choosing where to install ground instruments

Not good at

- Alarms in hours or minutes: images are days apart
- Rapid failures with little warning movement
- Slopes in forest, snow, layover or shadow
- Results delivered late if nobody processes them routinely

05

PART 5

Case studies

Mud Creek, Xinmo and Joshimath, and national ground-motion services

CASE STUDY

Mud Creek, Big Sur, California, 2017

17 cm/yr
Average speed from at least 2009, seen by airborne and satellite InSAR

≈ 40 cm/yr
Speed in the very wet winter of 2017

20 May 2017
Catastrophic failure onto coastal Highway 1

5 million
Cubic metres of material; the road was closed for over a year

Eight years of InSAR showed a slow landslide that sped up every rainy season.

A shift from historic drought to record rainfall pushed it from slow sliding to collapse.

Lesson: a ‘stable’ slow landslide can fail. Watch for a change in its normal behaviour.

Sources: Handwerger et al. (2019), Scientific Reports 9, 1569; NASA. Data: NASA UAVSAR and Copernicus Sentinel-1.

CASE STUDY

Xinmo, Maoxian, Sichuan, China, 2017

24 June 2017
A rock avalanche buries Xinmo village

≈ 83
People dead or missing; 64 houses destroyed

45
Sentinel-1 images in the archive, October 2014 to 19 June 2017

5 days
Between the last image and the failure

Processed after the disaster, the archive showed the source area moving and accelerating in the months before.

The authors concluded that routine monitoring could have given a warning.

Lesson: the data existed, but nobody was processing it for this slope.

Sources: Intrieri et al. (2018), Landslides 15, 123–133; COMET. Data: Copernicus Sentinel-1.

CASE STUDY

Joshimath, Uttarakhand, India, 2023

≈ 9 cm
Slow sinking, April to November 2022

5.4 cm
Rapid sinking in 12 days, 27 December 2022 to 8 January 2023

2,180 m
Height of the crown of the sinking zone, near the Joshimath–Auli road

Sentinel-1
Radar data used by NRSC (ISRO), checked with Cartosat-2S images

The town is built on old landslide deposits; cracks spread through houses and roads.

InSAR showed where the ground was sinking and that it had sped up.

Lesson: results must reach planners quickly, with clear limits and a plan for communication.

Source: National Remote Sensing Centre (ISRO), preliminary report, January 2023. Data: Copernicus Sentinel-1.

CASE STUDY

From one slope to a whole country

European Ground Motion Service

Copernicus, since 2022: Sentinel-1 velocities and time series for all of Europe, free, updated every year, including vertical and east–west products.

InSAR Norway

National Sentinel-1 ground-motion map since 2018, used to screen unstable rock slopes above fjords and valleys.

NISAR and India

NASA–ISRO L-band satellite, launched 30 July 2025, with free data: better coherence on forested Himalayan and Western Ghats slopes.

06

PART 6

Engineering practice

Uses, field instruments, limits and exercises

ENGINEERING

Where engineers use landslide InSAR

Roads and railways

Screen mountain corridors; find the creeping sections before they crack the road.

Dams and reservoirs

Watch the reservoir slopes as the water level changes.

Open-pit mines

Satellite screening plus ground-based radar on the active pit walls.

Pipelines and power lines

Long linear assets crossing many slopes: a wide-area method fits.

Towns on old landslides

Building damage, drainage and land-use planning, as at Joshimath.

After an earthquake or storm

Map new movement quickly and check whether remediation is working.

ENGINEERING

InSAR works best with field instruments

Instrument	What it adds to InSAR
GNSS station	Continuous 3D motion at one point: checks the InSAR line of sight
Inclinometer	Depth of the slip surface, from a borehole
Extensometer, crack meter	Opening of cracks at the head scarp, in real time
Piezometer	Groundwater pressure: the usual trigger
Rain gauge	Links rainfall to the speed-ups in the time series
Ground-based radar, total station	Readings every few minutes, for alarms

ENGINEERING

Limits and good practice

Limits

- Slow movements only (below about a metre a year)
- One direction; north–south motion nearly invisible
- Layover, shadow and steep slopes
- Forest, snow and seasons break coherence

Good practice

- Process both passes; compute C for every point
- Use a stable reference area nearby
- Do not project where |C| is below about 0.3
- Compare with rainfall, geology and field evidence

EXERCISES

Try these

01 A west-facing 30° slope moves 50 mm/yr downslope. Ascending C = +0.15, descending C = −0.93. What does each pass measure, and which would you use?

02 Why not simply divide the ascending value by 0.15 to get the downslope speed?

03 Sentinel-1 has a 12-day revisit. What is the largest velocity difference between neighbouring pixels it can measure? What would you do for a landslide moving 1 m/yr?

04 1 / velocity readings are 2.0, 1.4 and 0.8 days per mm on days 0, 30 and 60. If the trend continues, when is failure expected?

EXERCISES

Answers

01 Ascending: 50 × 0.15 ≈ **+7 mm/yr** (towards the satellite, close to the noise). Descending: 50 × −0.93 ≈ **−46 mm/yr**. Use descending.

02 Noise is divided by 0.15 too: ±2 mm/yr becomes ±13 mm/yr, and small errors in slope or aspect change C a lot.

03 λ/4 = 1.4 cm per 12 days ≈ **0.4 m/yr**. For 1 m/yr: 6-day pairs (≈ 0.8 m/yr), L-band such as NISAR (≈ 1.8 m/yr), or offset tracking.

04 The line falls 0.6 every 30 days, 0.02 per day, so it reaches zero on **day 100**: about 40 days after the last reading.

RECAP

Key points

01 InSAR measures slow landslides, millimetres to about a metre a year, over whole regions.

02 It sees only the line-of-sight part of the motion: slope direction decides how much.

03 Always use ascending and descending; mask layover and shadow; project only where $|C| \geq 0.3$.

04 Time series link movement to rainfall; acceleration and inverse velocity warn of failure.

05 Use InSAR to find and rank moving slopes, then confirm and monitor them on the ground.

FURTHER READING

References

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Thank you. Questions?

Figures and the animation were made for this lecture. Velocity maps, time series and the failure example are simulated with Sentinel-1 wavelength and geometry, and labelled as such; sensitivity values are computed for a 30° slope at 39° incidence. Velocity classes after Cruden and Varnes (1996). Case-study facts: Handwerger et al. (2019) and NASA (Mud Creek); Intrieri et al. (2018) and COMET (Xinmo); NRSC, ISRO (Joshimath). Sentinel-1 data: EU Copernicus programme / ESA. Full references on the previous slide.